SUMMER INTERNSHIP PROJECT REPORT ON

STUDY AND ANALYSIS OF THE POWER GENERATION MARKET

AT

GRAND VIEW RESEARCH, INC.

IN PARTIAL FULFILMENT OF THE REQUIREMENT FOR

THE AWARD OF A DEGREE OF

MASTER IN BUSINESS ADMINISTRATION (MBA)

BY

TARANPREET KAUR BHULLAR

MBA-III (MARKETING)

BATCH (2021-2023)

UNDER THE GUIDANCE OF

DR SHRIKANT WAGHULKAR

SUBMITTED TO

SAVITRIBAI PHULE PUNE UNIVERSITY

THROUGH

GRAMODAYA TRUST'S RAJEEV BUSINESS SCHOOL

PUNE, MAHARASHTRACOMPANY CERTIFICATE

GRAND VIEW RESEARCH

CIN NO. – U74900PN2014PTC152096

INTERNSHIP LETTER

18th January, 2023

To,
Taranpreet Kaur Bhullar
Market Research Intern
Intern ID: IN397

This is to certify that **Taranpreet Kaur Bhullar** has done her internship at **Grand View Research, Inc. from 17th May 2022 to 17th Aug 2022.**

Grand View Research (India) Pvt. Ltd.

She has worked on the project **"A study on market analysis of report on Power Generation Industry"** during her internship period.

During this internship she has demonstrated her skills with the best of her capabilities. Her performance met our expectations and she was able to complete the assigned tasks during this internship period.

We wish her all the best for her upcoming career.

Ashish Shenoy,
Head Of Human Resources.

*This is a computer generated letter and does not require any signature

705, A Wing, 7th Floor Amar Business Zone, Veerbhadra Nagar, Baner, Pune, Maharashtra 411045

CERTIFICATE

"To whom it may concern"

This is to certify that Miss Taranpreet Kaur Bhullar is a bona fide student of this institute enrolled in the MBA program.

She has completed her summer project titled A Study on Market Analysis of a Report on the Power Generation Market at Grand View Research, Inc., for partial fulfilment of the Ramachandran International Institute of Management, affiliated with Savitribai Phule Pune University.

We wish her good luck and best wishes for a bright future.

Internal Project Guide Prof. Dr Shrikant Waghulkar	Director Dr. Ajeet Sane

DECLARATION

I, Taranpreet Kaur Bhullar, hereby declare that the presented Summer Internship Report titled "A study on market analysis of the power generation market at Grand View Research, Inc." is uniquely prepared by me after the completion of the internship with Grand View Research, Inc. at Baner, Pune.

I also confirm that the report is only prepared for my academic requirement and not for any other purpose.

Place: Pune

Student's Signature:

Taranpreet Kaur Bhullar

ACKNOWLEDGEMENT

The success and outcome of this project required a lot of guidance and assistance from many people, and I am extremely privileged to have gotten this throughout the completion of my projects. All that I have done is only due to such supervision and assistance, and I would not forget to thank them.

I would like to convey my heartfelt gratitude to Mr Nilesh Patankar, Director at GVR, for his tremendous support and assistance in the completion of my project. I would also like to thank our institute for providing me with this wonderful opportunity to work on a project with the topic of studying and analysing the power generation market. The completion of the project would not have been possible without their help and insights.

I also owe my deep gratitude to my internal project guide for providing me with all the support and guidance, which made me complete the project. I chose this moment to acknowledge his contribution gratefully.

Last but not least, I'm thankful to all my teachers and friends who have always been helping and encouraging me throughout the year.

Sincerely,

Taranpreet Kaur Bhullar

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1 Introduction

1.1 Part 1

1.1.1 Why this research

Power generation is one of the most important utilities in any modern community. The power generation industry has two key tasks: producing power and distributing it within a predefined network. Both these aspects are being researched heavily in the present times due to their significant direct impact on human life. Thus, gaining new investment in order to improve generation and distribution processes and hence come up with progressive designs for the overall electric power system (EPS) is likely to remain a key aim of the global power generation market in the coming years. The global power generation market is anticipated to grow significantly in the coming years. And this factor makes it one of the most important researched topics in today's time.

1.1.2 Origin of this research

At Grand View Research, we worked and studied different market domains with many sub-categories, which involved Aerospace and Defence, Semiconductors, FMCG, Healthcare, IT, Electrical, Energy and Power Generation, etc. The Energy and Power Generation domain was among the most demanded research sectors owing to various climate change initiatives emphasised globally. Reports, at Grand View Research, take a closer look at various markets and focus on key markets. Such a holistic perspective on market dynamics, trends, supply, and demand that deep-dives into the market and segments it on multiple levels and provides key insights on a macro as well as micro-level, helped to choose this sector for research.

1.1.3 Significance of this research

The Indian marketing research industry has been growing by more than 30 per cent over the past three years. The spending on marketing research activities is presently estimated at $35 million. There are about 200 firms that provide marketing research services, of which eight to 10 firms are large and have nationwide capabilities. The torrid pace of growth is expected to stabilise at around

20 per cent per year. Since the total spending for research comprises only one per cent of total spending on all marketing services (advertising, marketing research, public relations, etc.), researchers believe the market holds tremendous potential.

Demand for research in this area of the power generation sector is inevitably followed. Recognising the need, several large marketing research companies have set up groups within their organisations to cater to this burgeoning market. The other major growth area has been in media services. Energy and Power generation are the upcoming markets gaining demand globally. Energy is an important material foundation necessary for the development of the national economy and people's living standards. It is also the basic material guarantee necessary for human production and life. In the long run, the production and consumption of energy should meet the requirements of sustainable development. Since research has always been the direction for recognising initiatives and strategies, it will play a significant role in estimating the power generation, future needs, new investment requirements, and forecast reports. The growing worldwide energy crisis has stimulated the need for sustainable energy and power generation, and research to analyse the demand.

The growing worldwide energy crisis has stimulated the need for sustainable energy and power generation. Various conventional energy sources, such as fossil fuels, are increasingly being replaced by renewable sources, such as wind energy and solar energy. These renewable sources of energy not only produce energy most efficiently but also have a minimal adverse environmental impact. Numerous governments throughout the world have framed various regulations regarding the reduction of greenhouse gas emissions.

1.1.4 Scope of the Market

For the scope of the market, this market is segmented into power generation by source, type, application, grid type, and geography.

1.1.4.1 Segmentation by Source

The power generation market is segmented by source of energy into

- Conventional/Non-Renewable Source
- Renewable Source

The conventional/non-renewable source market was the largest segment of the power generation market by source of energy, accounting for 68.4% of the total market in 2022. The renewable source market is expected to be the fastest-growing segment in the power generation market, segmented by source of energy, going forward at a CAGR of 12.1% during the 2021-2026 period.

1.1.4.2 Segmentation by Type

By power generation, the market is segmented into the following:

- Hydroelectricity
- Fossil Fuel Electricity
- Nuclear Electricity
- Solar Electricity
- Wind Electricity
- Geothermal Electricity
- Biomass Electricity
- Other Electricity

The fossil fuel electric power generation market was the largest segment of the power generation market by type, accounting for 56.4% of the total market in 2022. The solar electric power generation market is expected to be the fastest-growing segment in the power generation market, segmented by type, going forward at a CAGR of 23.6% during the 2021-2026 period.

The power generation mix is heavily skewed toward coal, with a major contribution, because of cheaper domestic fuel availability. Its share in the mix has increased over the years due to constraints in adding other conventional generation sources—hydro, nuclear, and gas. Coal thermal power plants are the most widely used thermal power plant and are expected to have the largest share in the electricity power (global) supply, in 2025, in the world.

1.1.4.3 Segmentation by Application

Based on application, the market is segmented into Commercial, Industrial, Residential and Transportation. Industrial power generation refers to applications in sectors such as manufacturing, food and beverage, chemicals, healthcare, etc. With the escalating call for a secure, sufficient and sustainable energy supply, the industrial segment is predicted to grow at a good pace in the upcoming years.

The residential segment was the largest segment of the power generation market by end-user, accounting for 35.0% of the total market in 2022. The transportation segment is expected to be the fastest-growing segment in the power generation market, which is segmented by end-user.

1.1.4.4 Segmentation by Grid Type

The power generation market is segmented by type of grid into

- Off Grid
- On Grid

The on-grid market was the largest segment of the power generation market by type of grid, accounting for 99.5% of the total market in 2022. The off-grid market is expected to be the fastest-growing segment in the power generation market, which is segmented by type of grid.

1.1.4.5 Geographical Segmentation

On the basis of geographical analysis, the global power generation market is segmented into five major regions, including North America, Europe, Asia Pacific, Latin America and the Middle East & Africa region.

Regionally, the Asia-Pacific is one of the major contributors to the global power generation market. Asia-Pacific is home to more than 50% of the global population and 60% of the large cities, and in the future, the continent will face increasing demand for power as millions of new customers are gaining access to electricity, with rapid population growth and industrialisation. Growing call for advanced power generation and distribution infrastructure in Southeast Asian countries and proactive government support to improve power generation and distribution networks in dynamic economies such as India and China are the main drivers of the power generation business of the Asia Pacific. The steady development of the renewable energy sector is likely to be the key factor for growth in the future.

North America has significant potential in terms of wind and solar energy production. Government regulations aimed at reducing greenhouse gas emissions have improved the use of renewable sources to generate electricity, resulting in a huge boost in the decentralised power generation market. Federal tax reform in the United States provides incentives to improve the energy efficiency of power systems to reduce overall electricity cost and carbon emissions.

The rest of the world markets, including Latin America (Brazil, Mexico, Argentina, and Chile) and the Middle East (UAE, Israel, South Africa, Ethiopia, Kenya, Egypt, Sudan and the rest of ME), are also predicted to have great potential. Companies operating here are in the developing stage and aspiring to become global with the help of developed economies and technologies.

1.1.4.6 Competitive Landscape

The organisation of the electrical sector of a country or region varies depending on the economic system of the country. In some places, all electric power generation, transmission and distribution is provided by a government-controlled organisation. Other regions have private or investor-owned utility companies, city or municipally owned companies, cooperatively owned by their own customers, or combinations. Generation, transmission and distribution may be offered by a single company, or different organisations may provide each of these portions of the system. Not everyone has access to grid electricity. About 840 million people (mostly in Africa) had no access in 2017, down from 1.2 billion in 2010.

The Power Market is fragmented. Some of the major players in the market

- ABB
- John Wood Group plc
- Alstom SA
- Siemens AG
- Wartsila
- TEPCO
- E.ON SE
- Huaneng Power International Inc.
- Southern Company
- Sinohydro Corporation
- Suzlon Group
- United Power
- Datang International Power
- State Grid Corporation of China
- Électricité de France S.A.

1.1.5 Utility: Who is going to benefit

Market analysis aims to assess the market for a specific good or service and forecast how the target market will react. This includes obtaining information for market segmentation and product

diversification, which are used to alter advertising campaigns or identify the most important aspects to clients. To finish the market research process, a business performs a number of tasks. Depending on the market sector under consideration, data is gathered. The company analyses and comprehends the generated data to determine whether there are any trends or pertinent data points that can be used in decision-making.

Grand View Research has grown at an incredible rate because it consistently innovates its research methods and techniques. Both personally and organizationally, this initiative is important. Grand View Research is attempting to expand its company into new verticals, so it must outline trends and research other markets. From Grand View Research's perspective, this initiative is a crucial step toward growing the consultancy business and providing market intelligence and analyses for different markets.

1.1.6 Objectives

- To study and analyse various factors driving and inhibiting the global power generation market.
- To analyse the impact of drivers and restraints on the power generation market growth for the market forecast.
- To study and present the analytical depiction of the power generation market, along with the current trends and future estimations
- To showcase the opportunities available in the market, which may act as future drivers for the market
- To provide a detailed analysis based on competitive intensity and how the competition will take shape in the coming years.

1.2 Part 2

1.2.1 Market Size

The global power generation market grew from $1,776.61 billion in 2021 to $1,913.39 billion in 2022 at a compound annual growth rate (CAGR) of 7.7%. The Russia-Ukraine war disrupted the chances of global economic recovery from the COVID-19 pandemic, at least in the short term. The war between these two countries has led to economic sanctions on multiple countries, a surge in commodity prices, and supply chain disruptions, affecting many markets across the globe. The power generation market is expected to grow to $2,543.23 billion in 2027 at a compound annual growth rate (CAGR) of 7.4%.

According to the World Energy Outlook 2019, global electricity demand increases at 2.1% per year to 2040, twice the rate of primary energy demand in the Stated Policies Scenario. Also, electrification across major end-use industries is expected to double the electricity demand by 2050. Furthermore, it is projected that renewable generation will account for more than 50% of the total power generation post-2035. Renewable electricity generation has surged by almost 3% owing to the new wind and solar PV projects completed over the past year, and renewables are generally dispatched before other sources of electricity. Also, with the rapid technological advancement, an increasing number of countries, companies, and regions are embracing sustainable energy generation, and the landscape is rapidly evolving. There have been several advancements in power generation technologies. One such example is the incorporation of machine learning capabilities of Artificial Intelligence (AI) with microgrid controllers, and such advancements will fuel the demand for power generation in the forecast period.

For instance, according to the BP Statistical Review of World Energy 2022, the primary energy consumption in the Asia-Pacific region increased from 220,48 exajoules in 2013 to 272.45 exajoules in 2021. China currently is the largest producer of solar and hydro energy in the APAC region. China produced more than 170 GW of solar electricity in the year 2018, and 1,302.00 TWh of hydro power plants in 2019, as per the report by the International Hydropower Association.

As of 2021, China has the most significant installed capacity globally. According to China's National Energy Administration, the country's renewable installed capacity reached 1.063 terawatts by the end of 2021, accounting for about 44.8% of the total installed generation capacity.

For instance, in 2022, the government of China announced that it plans to build 450 gigawatts of solar and wind energy power plants in the Gobi desert regions to achieve the renewable energy target by 2030.

According to the International Energy Agency (IEA), India is the third-largest energy-consuming country globally, which directly aids the power market in the nation. Over the coming years, millions of Indian households will buy new appliances and air-conditioners, and to support the growth, the nation will have to add a large amount of power systems in the forecast period. In addition, the Government of India is investing heavily in renewable energy to reduce carbon emissions, undertaking various large-scale sustainable power projects, and promoting green energy heavily. As of 2021, India's installed renewable energy capacity stood at 147 GW, and the country is targeting about 450 Gigawatt (GW of installed renewable energy capacity by 2030, which will directly aid the market.

1.2.2 Market Dynamics

1.2.2.1 Market Drivers

The growing world population is the main driver of the global power generation market. With the help of the rapid urbanisation of emerging regions, the increase in the number of consumers has led to a steady increase in electricity demand. This has forced governments around the world to take proactive steps to ensure a stable power supply in their jurisdiction, resulting in a steady stream of investments in the power generation sector.

Expansion in the historical period was the result of higher electricity consumption in developed and developing countries, government initiatives for energy infrastructure, economies, and improved energy technologies for the production of electricity. In the future, escalated use of electricity in transportation and the expansion of the world economy will drive the market boom.

The rising environmental concerns and government policies that provide incentives and tax benefits for solar panel installation, as well as the prohibitive cost of grid expansion, are projected to be important market drivers. In addition, technological advancements that have resulted in lower solar panel manufacturing costs and increased efficiency have been crucial contributors to the expansion of the distributed energy generation market. However, in the Asia-Pacific area, a lack of popular understanding and interest in the technology has impeded market expansion.

The manufacturers across Europe have been forced to align their efforts toward sustainable energy installations due to rigorous energy efficiency standards. The leading market companies, for example, are producing green transformers and microgrid infrastructure, which has had a favourable impact on the business environment. The current electrification regime, along with an increasing focus on decentralised energy generation, will bolster the market for distributed power generation.

1.2.2.2 Market Restraints and Challenges

Factors that might hamper future power generation market expansion include rising interest rates and environmental regulations. The biggest obstacle to market expansion is declining investment in different parts of the energy sector.

The existing electric generation systems, especially in developed countries, rely on an ageing infrastructure and struggle to meet the growing electricity demand. This ageing infrastructure requires replacement and the upgrading of existing systems, thus acting as a significant inhibitor for the market. For example, as of 2017, about 90% of electricity generated in the USA was generated by power plants that were 28 years old on average, and the existing nuclear power plant fleet is expected to retire between 2030 and 2040, causing a substantial loss of zero-emitting power plants. In addition, 35-48% of transmission and distribution (T&D) assets of power generation in the USA require replacement. This is particularly relevant for OECD countries such as the USA, the UK and Germany, where around 50% of the current grid infrastructure will reach 40 years of age by 2035.

1.2.2.3 Market Opportunities

The growing call for smart grids is likely to be the key feature of the development of the global power generation market in the coming years. Smart grid technologies are becoming more popular around the world due to the great advantage they offer in terms of efficiency and reliability. Smart grids are already in use in developed regions such as North America and Western Europe, and are predicted to enjoy greater adoption in developing countries in the coming years. This is a key advance for the worldwide power generation business, as it will improve almost every aspect of power distribution networks.

The rise in electric vehicles in developing countries will create significant opportunities for power generation companies in the market. For example, according to the International Energy Agency,

in 2020 the global electric car stock hit the 10 million mark, a 43% increase over 2019. Also, according to Bloomberg New Energy Finance (BNEF), EVs will hit 10% of global passenger vehicle sales by 2025, rising to 28% in 2030 and 58% in 2040.

1.2.2.4 Market Trends

Digital Technologies In Electric Power Generation

Digital technologies are increasingly making their way into the Power generation industry. Digital technologies such as smart grids, sensors and smart meters provide companies and customers with more accurate and real-time accounts of power usage. These technologies help improve productivity, efficiency, safety, compliance and reliability in power generation, which results in better asset management, planning, execution, and a faster level of service with higher customer satisfaction. For Instance, some major companies in the Power generation industry that adopted digital technologies include Duke Energy, Engie, National Grid and NextEra.

Advanced Ultra-Supercritical (AUSC) Technology For Thermal Power Plants:

Thermal power generation companies are investing in advanced ultra-supercritical (AUSC) pulverised coal technology to increase plant operating efficiency and reduce carbon emissions. AUSC operates with temperatures in the range of about 700-750°C. This technology also operates with efficiency greater than 50% and significantly reduces carbon emissions at the source. Many government-owned and private power producers are investing in these technologies to enhance power plants' efficiency and reduce carbon emissions. For instance, in September 2019, the US Department of Energy invested $56 million in coal technology projects; out of this $ 56 million, $5 million was allocated to advanced ultra-supercritical (AUSC) power plants. These investments are focused on using advanced materials in power generation activities to increase the efficiency of the plant and its environmental performance.

1.2.3 Market Reforms

The business model behind the electric utility has changed over the years, playing a vital role in shaping the electricity industry into what it is today, from generation, transmission, distribution, to the final local retailing. This has occurred prominently since the reform of the electricity supply industry in England and Wales in 1990.

United States: The Federal Energy Regulatory Commission (FERC) made a series of decisions that were intended to open the U.S. wholesale power market to new players, with the hope that spurring competition would save consumers $4 to $5 billion per year and encourage technical innovation in the industry. Steps were taken to give all market participants open access to existing interstate transmission lines. First was to vertically integrate electric utilities to functionally separate their transmission, power generation and marketing businesses to prevent self-dealing. Second was a system to provide all participants with timely access to information about available transmission capacity and prices.

The FERC also endorsed the concept of appointing independent system operators (ISOs) to manage the electric power grid - a function that was traditionally the responsibility of vertically integrated electric utility companies. The concept of an independent system operator evolved into that of organisations (RTOs). FERC's intention was that all U.S. companies owning interstate electric transmission lines would place those facilities under the control of an RTO.

Other countries: In some countries, wholesale electricity markets operate, with generators and retailers trading electricity in a similar manner to shares and currency. As deregulation continues further, utilities are driven to sell their assets as the energy market follows in line with the gas market in the use of the futures and spot markets and other financial arrangements. Even globalisation with foreign purchases is taking place. One such purchase was when the UK's National Grid, the largest private electric utility in the world, bought several electric utilities in New England for $3.2 billion. Domestically, local electric and gas firms have merged operations as they saw the advantages of joint affiliation, especially with the reduced cost of joint-metering.

Technological advances will take place in the competitive wholesale electric markets. Examples already being utilised include fuel cells used in space flight; aero derivative gas turbines used in jet aircraft; solar engineering and photovoltaic systems; offshore wind farms; and the communication advances spawned by the digital world, particularly with micro-processing, which aids in monitoring and dispatching.

1.3 Company Profile

Grand View Research is an India & US-based market research and consulting company, registered in the State of California and headquartered in San Francisco. The company provides syndicated

research reports, customised research reports, and consulting services. The Grand View Research database is used by the world's renowned academic institutions and Fortune 500 companies to understand the global and regional business environment. The database features thousands of statistics and in-depth analyses of 46 industries in 25 major countries worldwide.

The firms help clients make informed business decisions. Offer market intelligence studies ensuring relevant and fact-based research across a range of industries, including chemicals, materials, energy, healthcare, and technology. With a deep-seated understanding of many business environments, Grand View Research provides strategic objective insights.

Each year, Grand View Research completes more than 300 multi-country market studies aimed at rendering the following solutions:

➢ Syndicated/published research

Based on internal analysis involves keeping track of the ever-changing market trends**,** Identification of key problem areas and rendering solutions catering to a wide array of clients, Identification of subject based on interaction with key opinion leaders and industry experts, and undertaking studies by tracking strategic initiatives undertaken by major market participants. Undertaking studies by tracking product movement and the pipeline.

➢ Customisation/requirement-based studies

Studies based on published research market, intelligence studies designed or modified in compliance with sustainability strategies, target studies For instance, product-specific or regional/country-specific studies, and recommendation-based studies.

➢ Due diligence/consulting

Gives information about product benchmarking and position studies, market entry strategies, patent portfolio and IPR-based studies, opportunity assessment, commercialisation techniques, pricing modules, technology modules, technology benchmarking, and long-term client engagement due to diligence/consulting.

Services provided by GVR are the following

- Consulting Services

Consulting and advisory services provide a comprehensive, research-based view that is crucial for building business intelligence and overcoming market challenges. With their broad areas of expertise, the team of consultants is united in their commitment to the firm's clients and passionate about their business goals, which helps foster quick and effective decision-making. Grand View Research also identifies new growth opportunities to help organisations successfully establish and expand their business.

- Industry Reports

Grand View Research provides off-the-shelf, syndicated market research studies, publishing over 240 reports each year, covering 45 industries, on a global as well as regional level. Track various industries, identify key markets, and understand key macro and micro-economic trends. Keeping pace with the business environment, we publish custom, syndicated market research studies.

- Custom Research Services

Market research is a multi-faceted discipline, where requirements may vary in line with the strategic needs of an organisation. At Grand View Research, they have teams of analysts and domain experts, focusing on specific research functions to ensure that their clients always get the right people to get the job done.

➢ **The company offers various research reports on various verticals, such as:**

- Consumer Goods
- Semiconductors & Electronics
- Specialty & Fine Chemicals
- Food & Beverages
- Advanced Materials
- Healthcare
- Technology
- Bulk Chemicals

➢ **Vision**

To be known globally as an information integrator & analyser, providing actionable market data, industry insights, and consulting services to enable companies to take strategic decisions and achieve optimum performance, growth, and revenue targets.

➢ **Mission**

Our mission is simple. We want to achieve 100% customer satisfaction and spearhead the industry through service excellence and innovative business solutions.

- **Core Values**
  - Service Integrity
  - Diversity & Employee Satisfaction
  - Client Satisfaction
  - Value image
  - Innovation
  - Building a Kaizen Process
  - Social Responsibility

The registered address of the company

Grand View Research

Office No. 705, A Wing, 7th Floor, Amar Business Zone, Veerbhadra Nagar, Baner,

Pune, MH, 411045 India

Phone: +91-895-638-0888

Email: info-apac@grandviewresearch.com

Swot Analysis

- **Strengths**
  - Constant market monitoring
  - Continuous product monitoring
  - Continuous Kol Interviewing
  - Product positioning analysis
  - Partner prospecting
  - Data mining & analytics
  - Opportunity identification
  - Variable mapping

- **Weakness**
  - Time Gap Makes Research Irrelevant
  - Cost Consideration
  - Problem of Rapid Change
  - Problem of Trust and Accuracy
  - It is not a Problem-Solving Technique but an Aid to solve the Problem

- Subjective or Biased Result

- **Opportunities**
  - Wide/open market
  - Can give solutions to other verticals other than those it is offering now

- **Threats**
  - Economic Change
  - Competitive Innovation
  - Regulation
  - Consumer Perception
  - Addressing Threats

CSR Activities

The company believes that it has an obligation to society. Grand View Research takes various initiatives to provide something to the people. They typically work for the betterment of the education of children from rural India and Orphans.

2 Literature Review

Global Energy Demand 2021-2022

Author- Nuel Dsousa

Global energy demand is set to increase by 4.6% by 2024, more than offsetting the 4% contraction in 2022 and pushing demand 0.5% above 2021 levels. Almost 70% of the projected increase in global energy demand is in emerging markets and developing economies, where demand is set to rise to 3.4% above 2021 levels. Energy use in advanced economies is on course to be 3% below pre-COVID levels. Electricity demand is due to increase by 4.5% in 2023 or over 1 000 TWh. This is almost five times greater than the decline in 2020, cementing electricity's share in final energy demand above 20%. Almost 80% of the projected increase in demand in 2023 is in emerging markets and developing economies, with the People's Republic of China alone accounting for half of global growth. The power sector leads the way, with its demand for renewables on course to expand by more than 8%, to reach 8 300 TWh, the largest year-on-year growth on record in absolute terms. The United States, the European Union and India and China are expected to generate over 900 TWh from solar PV and wind in 2021, the European Union around 580 TWh, and the United States 550 TWh. Together, they represent almost three-quarters of global solar PV and wind output.

Source: https://www.iea.org/reports/global-energy-review-2021

Energy efficiency assessment of fixed asset investment projects – A case study of a Shenzhen combined-cycle power plant

Author: Li Yingjiana, Yousif Abakrb Qiu Qia, You Xinkuia Zhou Jipingc, Department of Mechanical, Materials and Manufacturing Engineering, The University of Nottingham Malaysia Campus, Malaysia

Approximately three-fifths of the country's power generation capacity depends on fossil fuels, mainly based on the domestic coal reserves. In the past decade, due to the lower capital requirements, shorter construction period and higher efficiency, the natural gas generation capacity has also grown very rapidly in the overall power generation capacity. Due to the increasing demand

for electricity and the increasing impact of population growth, Malaysia, an industrialised country, is considering many options for its future energy generation approaches.

The assessment of fixed asset investment projects of energy conservation is based on a scientific and systematic approach to analyse the energy production, consumption and management and helps to analyse whether the use of energy is in line with the national and local industrial policies and whether the project design meets China′s energy saving design outlines and industry standards. The level of energy consumption is compared to the domestic and international advanced levels regarding the project energy use and process analysis, in all aspects of energy production and energy consumption, taking technically feasible, economically reasonable and environmentally and socially responsible measures and effective and rational use of energy, to reduce consumption and losses and to stop waste, while investigating if the project has a certain advanced and rational use of energy.

https://www.researchgate.net/publication/263968878_Study_of_energy_recovery_and_power_generation_from_alternative_energy_source.

Recent developments of solar energy in India: Perspectives, strategies and future goals

Department of Electrical Engineering, National Institute of Technology Silchar, Assam, India

Prof. Subhojit Daw Prashant, Prof. Kumar Tiwari, Prof. Arup Kumar Goswami, Prof. Manash Kumar Mishra

Power is the most paramount ingredient of infrastructure for growth in economics and the welfare of a nation. Development for sustaining the growth of the Indian economy in the existing infrastructure is crucial. The power sector of India is one of the largest and most expanded power sectors in the world. Due to the continuous increment in electricity demand day-by-day, the Indian power sector is facing some challenges to maintain the balance between power generation and demand, suffering from supply constraints and shortages in power. For maintaining the ratio of generation and demand of power, moving from conventional sources to non-conventional sources is not only an option, but it is a necessity. The importance of using solar as an energy source in India′s perspectives is not only to increase power generation, but also to expand energy reliability by considering the environmental, social, independent and financial beneficial properties. This

paper analyses the recent scenario, strategies, availability, future potential, policies and development of solar energy in the emerging Indian power sector.

https://www.sciencedirect.com/science/article/abs/pii/S1364032116300570

The influence of the global electric power system on terrestrial biodiversity

Robert A. Holland, Kate Scott, Paolo Agnolucci, and Gail Taylor

Christopher B. Field, Stanford University, Stanford, CA, and approved October 30, 2019

The relationship between the size of individual components of the electric power sector and threat to biodiversity indicates that a shift to non-fossil sources, such as solar and wind, could reduce pressures on biodiversity both within the territory where power demand resides and along international supply chains. However, given the current levels of deployment of non-fossil sources of power, there is considerable uncertainty as to how the impacts of structural changes in the global electric power system will scale. Given the strong territorial link between demand and associated biodiversity impacts, the development of strong national governance around the electric power sector represents a clear route to mitigate threats to biodiversity associated with efforts to decarbonise society over the coming century. The influence of the global electric power system on terrestrial biodiversity

https://www.pnas.org/doi/10.1073/pnas.1909269116

A Literature Review on High Altitude Wind Power-Generation Systems

Author: Noel Dsouza

There has been a rising popularity with Alternate Energy Sources in the past few decades, with the Global Energy Demand and Supply Situation being brought into the spotlight. Although in a recent article published by the International Renewable Energy Agency (IRENA), future estimates of renewable energy contribution stand at Hydroelectric power leading the way with a 17% contribution to World Renewable Energy Consumption, Wind Power lags not too far behind at 14% contribution to Global Renewable Energy. Wind Power is primarily supplied through the use of Horizontal Axis Wind Turbines, among which the 3-blade design is the most frequently used. Although the wind turbine holds the rank of the most commonly used method, many studies show that higher wind speeds exist at altitudes above the reach of current wind turbines.

Power generation through renewable energy sources is the need of the hour, as we cannot be dependent on conventional energy sources to meet our needs. Solar energy is one of the most important renewable energy sources used all over the world [22]. Solar panels make use of the solar radiation reaching to the Earth and help in generating electric power with the help of solar radiation [23]. This power is utilised by us in our households, industries, offices, schools, colleges, etc.

https://www.ijert.org/a-literature-review-on-different-renewable-energy-resources

Power Markets Database1 Literature Review

By the Economic Modelling, Analytics and Research unit within the Sector Economics and Development Impact department of the International Finance Corporation

Electricity markets are relatively unique when compared to other markets due to the low demand elasticity for electricity, the specificity of competition resulting from a lack of product differentiation, and the quasi-public good nature of electricity provision, among others. Historically, power markets around the world began as vertically integrated monopolies with one utility, often state-owned, controlling all segments of the electricity value chain – generation, transmission and distribution. However, like most monopolies, vertically integrated utilities are usually characterised by inefficiencies resulting in low access rates, unreliable supply, and poor financial performance. As a result, power markets have evolved over time in most countries. Starting from the 1980's, many countries in Europe and the developing world adopted policy measures aimed at 1. The Power Markets Database was prepared by the Economic Modelling, Analytics and Research unit within the Sector Economics and Development Impact department of the International Finance Corporation.

https://www.worldbank.org/content/dam/Documents/Power-Markets-Database-Literature-Review.pdf

A Systematic Literature Review on Performance Evaluation of Power Systems from the Perspective of Sustainability

Sustainability is a comprehensive concept that integrates at least three dimensions of environment, economy and society. The power system is the primary source of greenhouse gas emissions, adversely impacting environmental sustainability. It also generates necessary energy supplies, which promote economic and social sustainable development. Based on the sustainability nature of power systems, this study puts forward an improved methodology, namely "Planning-Searching-Screening-Reporting-Reflecting" (PSSRR Cycle), to review the literature systematically on power system performance evaluation from a sustainability perspective over the past. Under the circumstances of advocating global low-carbon development, research on performance evaluation of power systems is shifting from the traditional model to the sustainability model, which expands many research fields in the future. Especially, in order to explore the sustainable transformation and development of the clean energy system from sustainability perspectives, the performance evaluation of the clean energy system is becoming a world research upsurge. Secondly, sustainable performance evaluation of power systems can adopt a combination of some management tools, such as the sustainable balanced scorecard (SBSC).

Future research should pay attention to the hybrid and application of the MCDM method and other methods. Finally, in terms of the geographical distribution of research on the performance evaluation of power systems, future research should focus on the sustainable development of power systems in underdeveloped regions such as Africa. Moreover, although solar power, hydroelectric power, wind power and even nuclear power are currently more common research objects, it can be noted that biomass and straw energy have less attention. Therefore, the research on sustainability performance evaluation of power systems can also be broadened into these new fields in the future.

https://www.frontiersin.org/articles/10.3389/fenvs.2022.925332/full

Changes in Energy Sector Strategies: A Literature Review

There is close cooperation between academia and the business environment to achieve sustainability in the energy sector. Energy sector transformation strategies aim to overcome the disadvantages caused by localisation and gain energy independence and diversification in energy generation. Therefore, there are multiple directions among those strategies. Currently, in most countries, the main challenge for the energy sector's strategies is to deal with energy security. The

implementation of SD induces changes both in strategy and technology. The strategies are based on the technological transition toward renewable energy sources (RES).

Business has always depended on and had an impact on the natural environment. However, through a proper strategy, it is possible to reverse or mitigate the negative influence of business. The energy sector, together with other economic areas' processes, negatively influences the environment, despite this sector being a condition of the development of civilisation. Energy generation worldwide is based on fossil fuel combustion, which especially increases negative climate change. Therefore, energy sector producers have been implementing strategies to change their business models, technologies, and sources of energy generation towards renewable energy sources (RES).

These transformation strategies are anchored in the sustainable development (SD) concept, but this idea is relatively new in business. With the increasing severity of climate change and widespread concern among the international communities, many countries have adjusted their energy strategies to increase the proportion of renewable energy (RE). RE has significantly developed over the past decade owing to the different forces. In 2010–2019, global non-hydro RE investment has reached approximately $2.6 trillion; investments in solar and wind energy reached approximately $1.3 trillion and $1 trillion, respectively. In 2018, the world's increased RE investment was approximately $332.1 billion. In 2017, RE power accounted for approximately 70% of the world's newly installed power capacity, and RE accounted for approximately 10.4% of the world's energy consumption. Of this total, RE electricity and hydropower accounted for 5.4% and 3.7%, respectively.

https://www.worldbank.org/content/dam/Documents/Power-Markets-Database-Literature-Review.pdf

A systematic bibliometric review of clean energy transition: Implications for low-carbon development

By Chengcheng Wang, Wei Cao

The transition from traditional energy towards clean energy remains the major challenge for the first half of the 21st century. We discuss the Energy Transition stream from two perspectives: obstacles in the clean energy transition and influencing factors on household energy transition. Current major challenges to clean energy transition include subsidies to traditional energy, high initial capital cost, high transaction cost, high financing risk, lack of price risk assessment, lack of clean technology, low market acceptance rate, and immature regulatory systems. Luthra et al. (2015) categorised 28 obstacles to the clean energy transition into seven dimensions: economy and finance, market, awareness and information, technology, ecology and geography, culture and behaviour, and political and government issues. For an in-depth look, the more challenging obstacles are ecological problems, consumers' lack of awareness of clean technology, inability to obtain solar radiation data, technical complexity, rehabilitation disputes and lack of political commitment.

https://journals.plos.org/plosone/article/authors?id=10.1371/journal.pone.0261091

A review of the clean energy transition

Wei Zhang, Binshuai Li, Rui Xue

Clean energy transition plays a crucial role in post-pandemic green recoveries and carbon neutrality. To advance understanding of the clean energy transition, this paper provides a systematic review of existing clean energy literature through a combination of bibliometric analysis techniques. Overall, there has been a surging trend of clean energy research since 2000, especially after 2016, when clean energy research experienced exponential growth. The future research agenda of clean energy awaits theoretical and practical exploration. We propose that the advancement of clean technology is at the heart of the clean energy transition and post-pandemic green recovery. Funding for clean energy transition is a critical challenge that needs innovative financial instruments and policy support. Thus, green bond markets, carbon taxes and emission trading system (ETS) need in-depth investigation. With more disruptive financing tools available, such as crowdfunding, efforts from enterprises and individuals also deserve more attention. In addition, international collaborations on clean energy transition projects are highly recommended. Intensive international collaborations and co-operations are of high importance to achieve low-carbon development. The completion of the global warming goal needs collective contributions from all countries around the world. A community of common destiny for all of humankind cannot be successfully built with efforts from only a small number of highly engaged countries. The current collaboration in clean energy research lacks worldwide collaborations in climate change actions. Therefore, it is highly recommended that all countries should shoulder their

responsibilities in climate change mitigation and adaptation, with steady growth of environmental investments and frequent collaborations with leading countries in climate change actions.

https://journals.plos.org/plosone/article/authors?id=10.1371/journal.pone.0261091

3 Research Methodology

Research methodology simply refers to the practical "how" of any given piece of research. More specifically, it's about how a researcher systematically designs a study to ensure valid and reliable results that address the research aims and objectives. For this research, an exploratory research method has been used. Exploratory research analysis focuses on the discovery of ideas and insights; it is applied to investigate the data and summarise the key insights. It will give us the basic understanding of our data, its distribution, null values and much more. That is why exploratory research is best suited as the beginning of a total research plan. It is most commonly used for further defining company issues, areas for potential growth, and alternative courses of action, and prioritising areas that require statistical research.

So, for this model, I've used secondary data, which was provided to me by Grand View Research Inc. itself. The research methodology entails an ideal mixture of primary and secondary initiatives. Key steps involved in the process are listed below:

3.1 Information Procurement

The stage involved the procurement of market data or related information via different sources & methodologies. GVR's Internal Database includes historic market databases, studies, internal audit reports & archives. They have a dedicated team of analysts updating & maintaining these databases, i.e. Primary Research and Secondary research. For this research, secondary sources (Company filings, Investors documents) & third-party perspectives were used, which included government statistics published by organisations. Third-party perspectives include the analysis of investor analyst reports, academic commentary, government quotes & wealth management publications.

3.2 Information Analysis

This step involved the analysis & mapping of all the information procured from the previous step. It also encompasses the analysis of data discrepancies observed across various data sources. Information procured from secondary and primary initiatives is then analysed by using the following tools/models:

a) Identifying variables and establishing the market impact
b) Establishing market trends

c) Analysing future opportunities and market penetration rates by understanding commercialisation, regional expansion, etc.
d) Analysing reimbursement/regulatory trends and changes in market dynamics to establish future growth.
e) Analysing sustainability strategies adhered to by market participants in an attempt to determine the future course of the market
f) Analysing historical market trends and imposing them on the current and future variables to determine the year-on-year trend
g) Understanding consumer procedure trends and the regulatory framework
h) Base numbers are established by analysing the company revenues and market share, and deriving market estimates via analysing parent and ancillary markets.

3.3 Market Formulation

The final step entails the placement of data points in appropriate market spaces in an attempt to deduce viable conclusions. The market dynamics and trends helped to drive the conclusion of the overall power generation market globally. Recent developments help determine the future aspects with respect to opportunities for growth in the market, which helps in strategies and decision-making for the investors.

3.4 Limitation

a) Every project does have some limitations in terms of the constraints faced during various stages of the study, e.g. data collection, analysis, etc.
b) Data collection
c) Scarce data is present about the company's financials
d) Analysis of large data
e) Difference in perspective – different people can have different perspectives on a particular topic, so we can expect the results to be different or biased.
f) Limited secondary data- since not many studies were conducted on this part, it was very difficult to get the desired data.

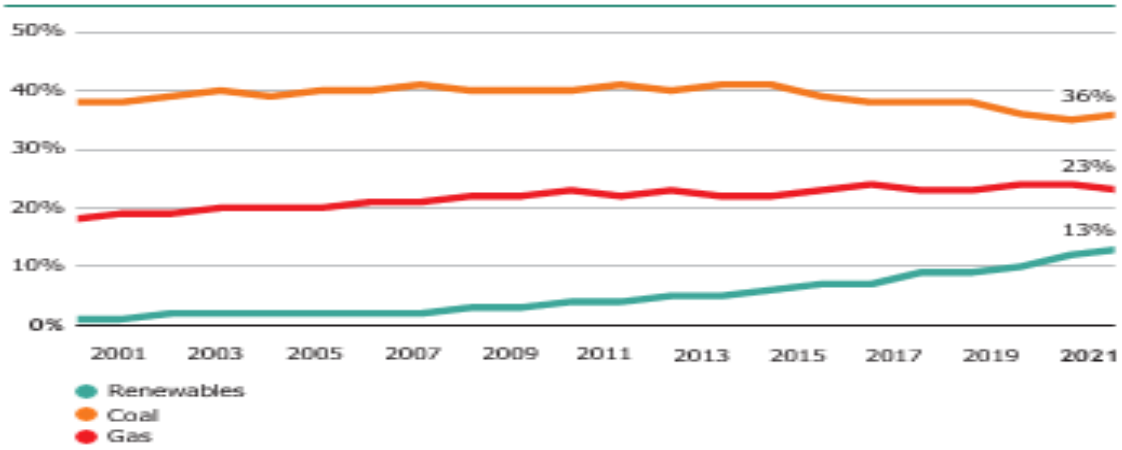
4 Data analysis

Primary Power Generation

Primary energy grew by 31 EJ in 2021, the largest increase in history and more than reversing the sharp decline seen in 2020. Primary energy in 2021 was 8 EJ above 2019 levels. The increase in primary energy in 2021 was driven by emerging economies, which increased by 13 EJ, with China expanding by 10 EJ. Since 2019, primary energy consumption in emerging economies \increased by 15 EJ, largely reflecting growth in China (13 EJ). In contrast, energy demand in developed economies in 2021 was 8 EJ below 2019 levels. The increase in primary energy between 2019 and 2021 was entirely driven by renewable energy sources. The level of fossil fuel energy consumption was unchanged between 2019 and 2021, with lower oil demand (-8 EJ) offset by higher natural gas (5 EJ) and coal (3 EJ) consumption.

Electricity generation increased by 6.2% in 2021 – similar to the strong bounce back seen in 2010 in the aftermath of the financial crisis (6.4%). Wind and solar reached a 10.2% share of power generation in2021, the first time wind and solar power have provided more than 10% of global power and surpassed the contribution ofnuclear energy. Coal remained the dominant fuel for power generation in 2021,with its share increasing to 36%, up from 35.1% in 2020. Natural gas in power generation increased by 2.6% in 2021, although its share decreased from 23.7% in 2020 to 22.9% in 2021.

The share of renewable in global power generation continued to increase



The share of renewables (excluding hydro) in global powergeneration continued its rising trend, driven by strong expansion in solar and wind energy. Renewables' share is almost 13% in 2021, higher thanthe share of nuclear energy (9.8%). Energy from coal increased slightly from35% to 36% in 2021 but remained below its 2019 level. In the last two years, renewables have grown more than the combined increasein coal and natural gas supplies.

India is the third-largest producer and consumer of electricity worldwide, with an installed power capacity of 408.71 GW as of October 31, 2022. As of October 31, 2022, India's installed renewable energy capacity (including hydro) stood at 165.94 GW, representing 40.6% of the overall installed power capacity. Solar energy is estimated to contribute 61.62 GW, followed by 41.84 GW from wind power, 10.70 GW from biomass, 4.92 GW from small hydropower, and 46.85 GW from hydropower. The non-hydro renewable energy capacity addition stood at 4.2 GW for the first three months of FY23 against 2.6 GW for the first three months of FY22. With electricity generation (including renewable sources) of 846.18 BU in India between April and September 2022, the country witnessed a growth of 10.67% YoY. According to data from the Ministry of Power, India's power consumption increased 1.64% YoY in October 2022 to 114.64 BU. The peak power demand in the country stood at 210.79 GW on June 9, 2022. All India actual PLF of thermal power plants stood at 68.24% in June 2022, compared with 67.92% in May 2022.

Electricity Generation on a regional basis (North America and Asia Pacific)

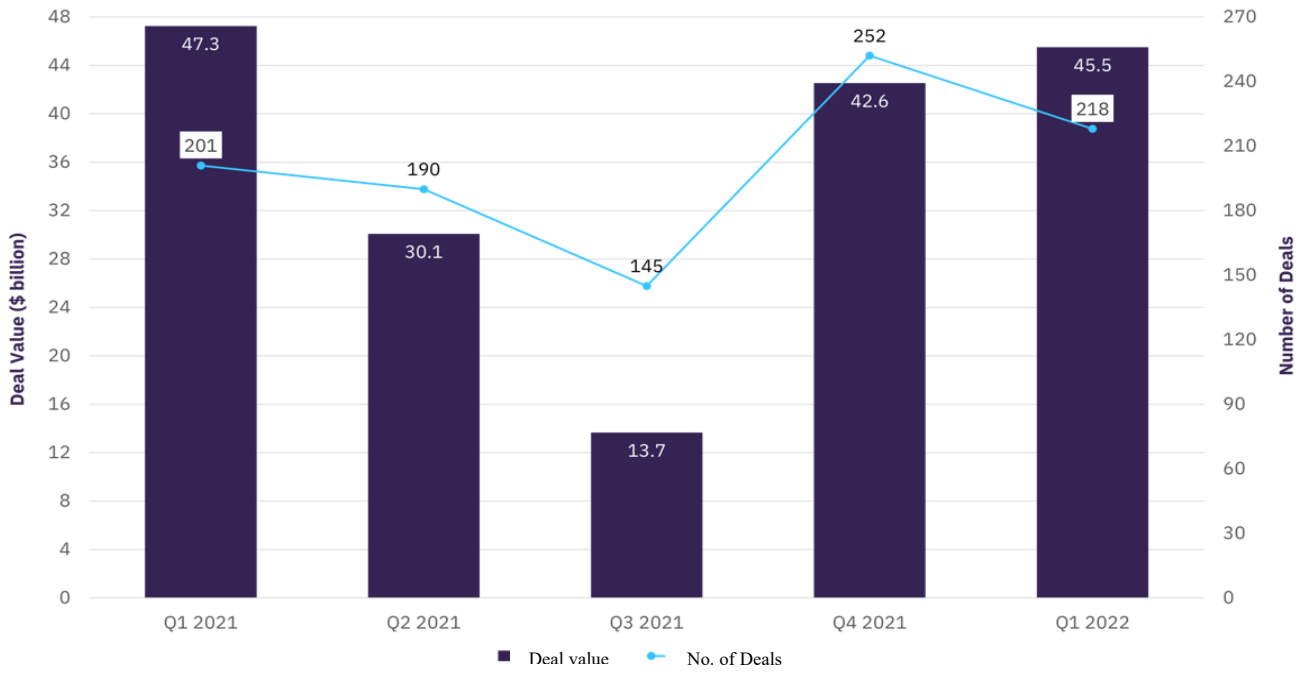
Electricity Generation*					Growth rate per annum		Share
	2018	2019	2020	**2021**	2021	2011-21	2021
Terawatt-hours							
Canada	655.8	650.8	649.1	**641.0**	-1.0%	♦	2.3%
Mexico	335.1	344.6	325.7	**336.0**	3.4%	1.4%	1.2%
US	4461.6	4411.2	4284.8	**4406.4**	3.1%	0.1%	15.5%
Total North America	**5452.5**	**5406.5**	**5259.7**	**5383.5**	**2.6%**	**0.2%**	**18.9%**
Australia	262.6	265.9	265.2	**267.5**	1.1%	0.4%	0.9%
Bangladesh	81.1	89.7	89.1	**97.9**	10.2%	8.3%	0.3%
China	7166.1	7503.4	7779.1	**8534.3**	10.0%	6.1%	30.0%
China Hong Kong SAR	36.5	36.9	35.2	**37.1**	5.7%	-0.5%	0.1%
India	1579.2	1622.1	1563.3	**1714.8**	10.0%	5.2%	6.0%
Indonesia	283.8	295.4	291.8	**309.4**	6.3%	5.4%	1.1%
Japan	1053.2	1025.8	997.0	**1019.7**	2.6%	-0.8%	3.6%
Malaysia	170.6	178.5	169.5	**177.2**	4.9%	3.4%	0.6%
New Zealand	44.7	45.1	44.5	**44.4**	♦	♦	0.2%
Pakistan	137.8	137.5	135.3	**152.1**	12.7%	4.2%	0.5%
Philippines	99.8	106.0	101.8	**108.3**	6.7%	4.6%	0.4%
Singapore	52.9	54.1	53.1	**55.8**	5.4%	1.9%	0.2%
South Korea	592.9	586.8	575.3	**600.4**	4.7%	1.5%	2.1%
Sri Lanka	16.2	16.8	16.8	**17.4**	3.7%	4.1%	0.1%
Taiwan	275.5	274.2	280.0	**290.9**	4.2%	1.4%	1.0%
Thailand	177.8	186.5	176.5	**176.3**	0.2%	1.4%	0.6%
Vietnam	209.2	227.4	235.4	**244.8**	4.3%	9.2%	0.9%
Other Asia Pacific	132.6	131.5	140.5	**146.1**	4.3%	6.6%	0.5%
Total Asia Pacific	**12372.4**	**12783.7**	**12949.3**	**13994.4**	**8.4%**	**4.7%**	**49.2%**
Total World	**26677.3**	**27036.6**	**26889.2**	**28466.3**	**6.2%**	**2.5%**	**100.0%**

Deal Analysis of Global Renewable Power Industry in Q1 2022

As per the Global Data Power database, the global renewable power industry registered 218 deals worth $45.5 billion in Q1 2022, as compared to 252 deals worth $42.6 billion in Q4 2021. On a year-on-year basis, the number of deals increased, and deal value decreased in Q1 2022, as compared to 201 deals worth $47.3 billion in Q1 2021. Asset transactions in the renewable power industry constituted the largest segment with 107 deals, which accounted for 49.1% of the total deals in Q1 2022, followed by mergers & acquisitions (M&A) with 51 deals representing 23.4%, PE/VC and equity offerings both with 19 deals each accounted for 8.7% share of the total deals.

Solar market registered a decrease in the number of deals and an increase in deal value with 132 deals worth $20.9 billion in Q1 2022, as compared to 166 deals worth $20.5 billion in Q4 2021. While the wind power market registered a decrease in the number of deals and a slight increase in deal value, with 72 deals worth $19.64 billion in Q1 2022, as compared to 89 deals worth $19.62 billion in Q4 2021. On the other hand, the hydro power market reported a decrease in the number of deals and an increase in deal value with 22 deals worth $3.1 billion in Q1 2022, as compared to 37 deals worth $1.9 billion in Q4 2021.

Europe leads global renewable deal announcements. Europe continues to remain a region that announced the maximum number of deals for the fifth quarter in a row and the largest market in terms of total deal value for the second quarter in a row. The region witnessed a decrease in the number of deals and deal value, with 90 deals worth $18.6 billion in Q1 2022, as compared to 113 deals worth $22 billion in Q4 2021. Europe is followed by the Asia Pacific region, which registered a slight decline in the number of deals and a huge increase in deal value with 61 deals worth $10.4 billion in Q1 2022, as compared to 64 deals worth $4.7 billion in Q4 2021. North America registered a decrease in the number of deals and an increase in deal value with 42 deals worth $13.4 billion in Q1 2022, as compared to 55 deals worth $11.6 billion in Q4 2021.

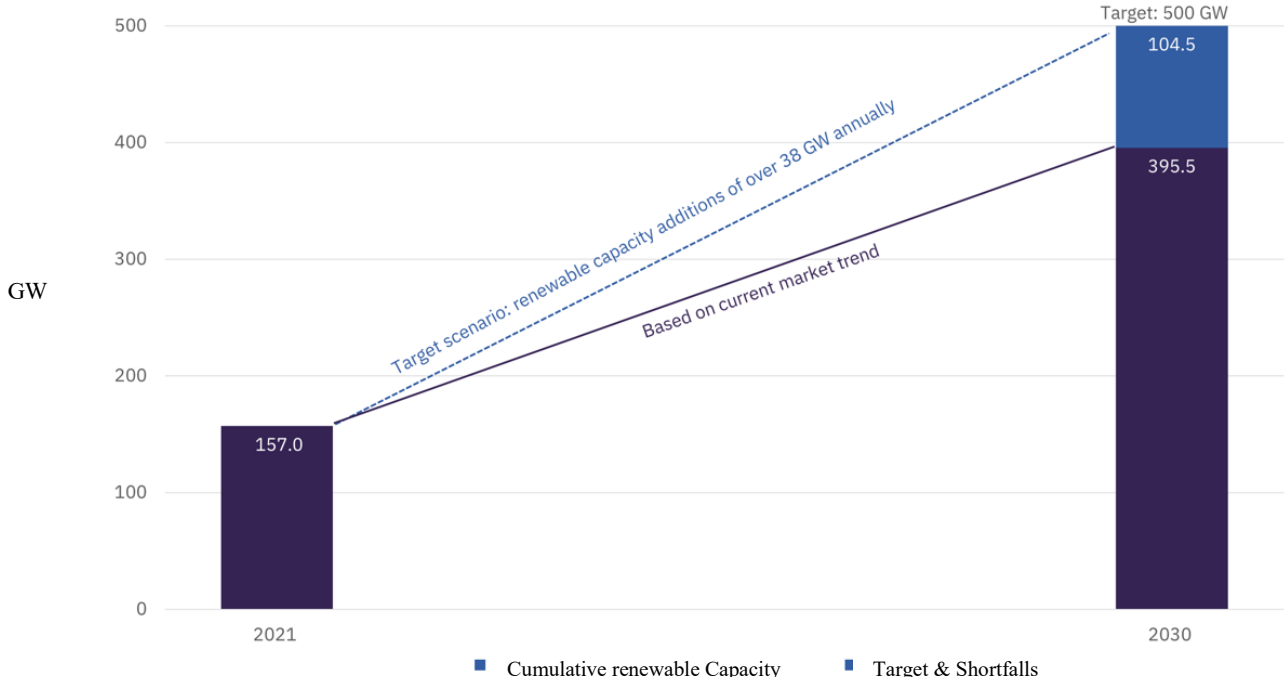


As per Global Data's Power database, India is the third-largest renewable power market (including total hydro) in terms of both annual additions and cumulative installed capacity in 2021. The country added 14.6 GW of renewable capacity in 2021, and its total renewable capacity stood at 157 GW. India is behind China (121 GW) and the US (38 GW) in terms of annual renewable capacity additions. It also stood at third place in terms of cumulative renewable capacity behind China (1,065 GW) and the US (379 GW) in 2021.

A renewable capacity addition in India is mainly led by solar PV, which installed 12.8 GW capacity and accounted for 87.4% of total renewable additions in 2021. The country improved its ranking to third from fifth position a year earlier with respect to PV capacity additions. Expansion of renewables (includes solar PV) in India in the last decade is mainly driven by the 175 GW renewable target by 2022, of which the target for solar is 100 GW. India is expected to achieve its 175 GW target, having added large hydro and pumped storage under its renewables definition; however, it will fail to achieve its 2022 solar PV target.

The solar PV market is key for India to achieve its renewable target and climate goals. Solar PV will play a key role in India's progress to achieve its renewable target for 2030 and long-term climate goals. In the COP26 Conference, India announced new targets which include increasing the share of non-fossil energy capacity to 500 GW and meeting 50% of the country's electricity needs through renewables by 2030. The target also includes achieving net zero emissions by 2070.

The country began to focus on rooftop solar, along with large ground-mounted solar parks, in achieving its solar-specific targets as well as renewable goals. India also intends to explore its offshore wind potential, along with increasing capacity additions in onshore wind. The current growth trend of annual renewable additions will not help the country achieve its 500 GW target by 2030. To achieve the 2030 target, India needs to add over 38 GW annually during 2022-2030.



5 Findings

This report, by Grand View Research, takes a closer look at the power generation industry, providing a holistic perspective on market dynamics, trends, supply, and demand. The report deep-dives into the market and segments it on multiple levels, and provides key insights on a macro as well as micro-level. The study aims to provide granular information regarding estimates and forecasts for key power generation sources, including renewable and non-renewable, as well as applications, including residential, commercial, and industrial. The report also provides the same level of information for key regional markets. This research report analyses various factors that have been driving and inhibiting the global power generation market in the past. Besides, the report analyses the impact of each driver and restraint on the market growth, which forms the base for the market forecast. The report also showcases the opportunities available in the market, which may act as future drivers for the market.

The growing global population is the key driver for the global power generation market. Aided by the rapid urbanisation in emerging regions, the rising number of consumers has resulted in a steady rise in the demand for electricity. This has forced governments across the world to take proactive steps to ensure a steady power supply in their jurisdiction, leading to a steady influx of investment into the power generation industry. This is likely to remain a leading driver for the global power generation market in the coming years. The rising demand for smart grids is likely to be a key feature of the development of the global power generation market in the coming years. Smart grid technologies are becoming popular across the world due to the significant benefits they provide in terms of efficiency and reliability. Smart grids are already in use in developed regions such as North America and Western Europe, and are likely to enjoy rising adoption in developing countries in the coming years. This is a key development for the global power generation market, as it will enhance almost every aspect of power distribution networks.

A large amount of land is required for utility-scale power-producing installations to be efficient. Many factors, such as the availability of solar radiation, can limit the selection of land for solar power generation. Due to the amount of land occupied by the power plant, solar systems installed on a large scale might have an influence on natural regions and biodiversity. As a result, land use is constraining the growth of the distributed energy generation market during the forecast

period. Factors that might hamper future power generation market expansion include rising interest rates and environmental regulations. The biggest obstacle to market expansion is declining investment in different parts of the energy sector. The existing electric generation systems, especially in developed countries, rely on an ageing infrastructure and struggle to meet the growing electricity demand. This ageing infrastructure requires replacement and the upgrading of existing systems, thus acting as a significant inhibitor for the market.

According to the International Energy Agency (IEA), renewables will have the fastest growth rate in the electricity sector, meeting approximately 30% of total power demand by 2023, up from 24% in 2017. Renewables are expected to account for 70% of global electricity generation growth during this period, led by solar PV, followed by wind, hydropower, and bio-energy, in that order. In 2017, global power sector investment in renewable energy generation totalled USD 750 billion. As these technologies gain traction for power generation, the use of batteries for off-grid and grid energy storage has increased, aided by battery cost reductions. All of these factors pose major challenges for key players in the market for advances in distributed energy generation.

It is found that the speculative movement of the corporate sector has been influenced to an expansive degree because of the recession in the economy. Yet, it is the fact that it has not subsided to influence the power demand but rather has affected the sources of funds for new projects. Furthermore, since the incubation time frame for power plants is huge, the speculators are presently more careful than some time recently. They explore completely before investing to know the free cash flow in this sector. More and more firms are exploring supernormal returns by entering this profitable sector of power generation, transmission or distribution and are willing to acknowledge the fluctuations in the estimation of their portfolio, which is part of this sector, while at about the same rate, the companies are willing to put resources into less risky ventures where fluctuations are limited.

The main conclusion of this study is that there are changes in the energy sector, but the pace of those changes is too slow in relation to the real actions. This disparity is caused by the imbalance between struggles to improve the quality of life and stop climate change, and especially the rise in levels. Energy sector transformation strategies aim to overcome the disadvantages caused by localisation and gain energy independence and diversification in energy generation. Therefore,

there are multiple directions among those strategies. There are different driving forces behind the strategic changes in the energy sector.

Recommendations

To take advantage of the opportunities, The Business Research Company recommends the power generation companies to use advanced ultra-supercritical (AUSC) technology for thermal power plants, invest in nuclear power generation, adopt digital technologies in electric power generation, explore power generation from waste, partner with local distributors, focus on localized power generation, set up solar power plants in deserts and oceans, provide competitively priced offerings, offer fixed priced product/services, leverage social media to maximize reach, focus on promoting green credentials and focus on educating people on smart use of electricity.

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